$$N_{MV}(n) = n \cdot (2m-1)$$

※ SVD

$$O(n^3)$$

$$= 21n^3 + O(n^2)$$

1) What is SVD?

2) What is reduced SVD?

3) How do you use it to approximate the matrix?

Set small singular values to 0

QR Factorisation

$$O(nm^2)$$

<u>In application</u>

$$\overset{k}{\begin{bmatrix} A_1 \end{bmatrix}} \begin{bmatrix} A_2^T \end{bmatrix} + \overset{k}{\begin{bmatrix} B_1 \end{bmatrix}} \begin{bmatrix} B_2^T \end{bmatrix} = \begin{bmatrix} A_1 B_1 \end{bmatrix} \begin{bmatrix} A_2^T \\ B_2^T \end{bmatrix}$$

$A \in R_k$ $\qquad$ $B \in R_k$ $\qquad$ $AB \in R_{2k}$

$$\approx \overset{k}{\begin{bmatrix} C_1 \end{bmatrix}} \begin{bmatrix} C_2^T \end{bmatrix}$$

Truncate to rank k

from rank $L \to k$?

4) Where would we need to reduce from [illegible]

~Approximation addition of rank k

→ Exercise

Truncation is not associative

$$M_1 + M_2 + M_3 \neq \underbrace{M_1 + (M_2 + M_3)}_{\text{This is pairwise truncation}}$$

Pairwise truncation is faster & has less computation time.

$$M_1 = M_2 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad M_3 = \begin{bmatrix} 0 & 0 \\ 0 & 1.5 \end{bmatrix}$$

$$M_1 + M_2 + M_3 = \begin{bmatrix} 2 & 0 \\ 0 & 1.5 \end{bmatrix}$$

Truncation → lowest pt in diagonal → make it 0

$$= \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$$

→ The truncation of matrix in $R(l, n, m)$ to rank k with $k < l$ requires fewer than

$c(l^2(n+m) + nml^3)$ operations

$6L(n+m)+22L$

for exam we should know how L^2, L^3 came from.

→ Lecture 3

1) What is H_p matrix?

Define & Structure

2) Asymptotic complexities of these matrix approximations.

a) $H_p \cdot v$ (Matrix & vector) – $O(n \log n)$

Should know how we got these.

Why $n \log n$

n is no of levels $\log n$ - cost for level

b) $N_{H \odot H}(p)$ (matrix Matrix multiplication)

$O(n \log^2 n)$

$(n \cdot \log n) \cdot \log n$

no of levels we go

cost for level

c) Complexities for other computations and how we got those.

In $H_p \cdot V$ $O(n \log n)$ this is for $k=1$

$k =$ Rank 1 if we have k rank

then $O(k.n \log n)$

d) $H_p \oplus H_p$

$$\begin{pmatrix} H & R \\ R & H \end{pmatrix} + \begin{pmatrix} H & R \\ R & H \end{pmatrix} = \begin{pmatrix} H+H & R+R \\ R+R & H+H \end{pmatrix}$$

$$O(\underline{\underline{k^2}} n \log n)$$

→ <u>Lecture 4</u>

1) Advantages & Disadvantages of using Taylor expansion to approximate and find error bounds.

2) Discuss about error bound.

$$\eta = \frac{\text{diam}(I_{left})}{2\,\text{dist}(I_{left}, I_{right})}$$

high k or low k

big diameter or small diameter

→ <u>Lecture 5</u>

1) Why do we need & use separable functions?

[illegible] & use Interpolation?

2) Why do we need & use interpolation?

Easier than Taylor exp-
-ansion & more
accurate.

to get rank k matrix

3) Why do we need Chebyshev polynomials?

4) What is a stability constant? [C_{stab}]
Why do we use it?

5) What is C_L (Lebesgue stability constant)?

→ <u>Lecture 6</u>

We need $\boxed{\text{diam}(\Omega_s) \leq \frac{4}{C_0} \text{ dist}(\Omega_s, \Omega_t)}$

rank : $k = (m+1)^d$

$k^{1/d} = m+1$

Admissibility condition

1) What is exponential convergence?

2) If we have separable approximation by interpolation what is the α compo-
-nent.

... exponential convergence

3) 2 assumptions for exponential [illegible]
or prerequisites

A: 4.10 & 4.11

① Admissibility condition

② Asymptotic smoothness

→ 5.1 to 5.3 → Next Lecture

→ Lecture 7

1) What is a cluster tree?

↳ Tree & cluster tree?

2) How do we partition in clusters?

↳ Geometry based and
cardinality based partitioning

3) What is the difference b/w Geometry based & cardinality based partitioning?

4) Why do we use bounding boxes for clusters?

5) What is resolution of cluster tree?

6) What is a block cluster tree?

→ 5.4 & 5.5 Next Lecture

→ 5.4 & -

※ Read proofs of theorems → they are subjects of oral Exams

Theorem 5.16 ※

→ Lecture 8

1) Proof of storage of $\mathcal{H}$-matrix

※ Theorem 5.16 ※

1. Step: storage of leaf-block $s \times t$

a) Adm (s,t): false

b) Adm (s,t): true

2. Step: Storage of $\mathcal{H}$ matrix

should be able to explain this proof in words

2) In Example 5.18 $C_{sp} \leq 6^d$

What is d?

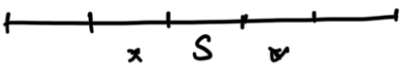
(d: spatial dimension)

— Why 6^d?

- In d dimensions → at most 3^d neighbours
- Each neighbour subdivides into 2^d

subclusters

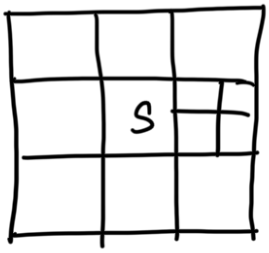
In 1D



$3^1 . 2^1 = 6$

* 2 non admissible clusters

In 2D



$s \times t$

$3^d . 2^d$

$3^2 . 2^2 = 36$

3) What is the Norm estimate for error?

Frobenius norm *

square root of sum of squared errors.

→ Next Lecture

6.1 - 6.3

Lecture 9

1) We computed things in lemma 6.1 i.e vector multiplications, where is the addition part?

↳ It is included in the #S

operations

* The work for matrix-matrix addition is the work done on the leaves.

- cost of adding inadmissible leaf with another inadmissible leaf

$s \times t$

- cost of adding 2 admissible leaves when we add them rank increases so we do truncation

  ↳ i.e QR decomposition

- Truncate to rank k

  in $O(ck^2(\#s, \#t))$

  c ↓ constant

2) How do we do matrix-matrix multiplication?

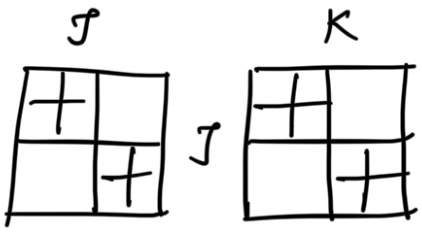
↳ We do recursively in the block structure

→ <u>Theorem 6.6</u>

3) Why 3 indices all of sudden

T_I, T_J, T_K

$L \because$



4) Where is $(\ )^2$ the square coming from in the theorem equation?

→ <u>Next Lecture</u>

<u>6.4, 6.5, 7.1</u>

<u>Lecture 10</u>

1) What are the 2 methods / possibilities to obtain inverse of H matrices?

- Schur complement
- Newton iterative

2) complexities for

LU decomposition $O(n \log^2 n)$

H-matrix inverse $O(n \log^2 n)$

But LU is faster. Why? How?

3) H-matrix multiplication : $O(k^2 n \log^2 n)$

k^2 here?

Why [illegible]

and even for other

arithmetic matrix operations.